

P3068R1: Allowing exception throwing in constant-evaluation

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Audience: EWG
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Motivation

Since adding the `constexpr` keyword in C++11, WG21 has gradually expanded the scope of language features for use in constant-evaluated code. At first users couldn't even use `if`, `else`, or loops. C++14 added them. C++17 added `constexpr` lambdas. C++20 finally added the ability to use allocation, `std::vector` and `std::string`. These improvements have been widely appreciated by many users, and they lead to simpler code that doesn't need to work around differences between normal and `constexpr` C++.

The last major language feature from C++ still not present in `constexpr` code is the ability to throw an exception. This absence forces library authors to use more intrusive error reporting mechanisms. One example would be the use of `std::expected`, `std::optional`. Another one is complete omission of error handling. This leaves users with long and confusing errors generated by the compiler.

Throwing exceptions in constant evaluated code is the preferred way of error reporting in the proposal adding Static Reflection for C++26¹. Some meta-functions can fail and allowing them to throw will significantly simplify reflection code.

Changes

- R0 → R1: Changed wording after consultation with Jens, added example to show exception can't leak via `std::exception_ptr`

Proposed changes in wording

7.7 Constant expressions [expr.const]

(5.25) a throw-expression ([expr.throw]), unless the lifetimes of the exception object and any implicit copies of exception object created by calls to `std::current_exception()` or `std::rethrow_exception()` end within the evaluation of E

Implementation experience

None in a C++ compiler. The author implemented this in simple AST walking scripting language which is how `constexpr` code is evaluated in most of C++ compilers.

The implementation strategy is usually registering a handler for specific exception types when `try` and `catch` blocks are found and unregistering when the `try` block is left.

Impact on existing code

This change shouldn't break any existing code as throwing exceptions without catching them is already an error and is used by various libraries² to improve compile-time errors.

The intent is to keep this useful mechanism intact. The proposed wording change will only modify behavior in cases where there is `try/catch` block present.

Only language changes in this paper

Intention of this proposal is to allow throwing any type of exception as long as it can be constructed in constant evaluated context. We don't propose making a magic type of a `constexpr` exception type to be inherited from.

This paper doesn't propose any library changes, but we think another paper should mark helper functions to be `constexpr`, namely: `std::current_exception`, `std::uncaught_exception`, and `std::rethrow_exception`.

¹ <https://wg21.link/P2996R1#error-handling-in-reflection>

² `libfmt` (<https://github.com/fmtlib/fmt/blob/master/include/fmt/format.h#L2245>)

`CTHASH` (<https://github.com/hanickadot/cthash/blob/main/include/cthash/sha2/sha512/t.hpp#L18>)

What is allowed and what's not?

Defining a new `constexpr` variable always creates a new constant evaluation context. `try/catch` blocks around such definition won't catch the exception thrown from inside of it and it will lead to a compile-time error. This behavior is similar to `constexpr` memory allocations, which can't leave the `constexpr` context.

```
constexpr auto just_error() {
    throw my_exception{"this is always an error"};
}

constexpr void foo() {
    try {
        auto v = just_error(); // OK
    } catch (my_exception) { }

    try {
        constexpr auto v = just_error(); // ERROR: constexpr variable creates new constant evaluation context
    } catch (my_exception) { }
}
```

Exceptions must be caught

```
constexpr unsigned divide(unsigned n, unsigned d) {
    if (d == 0u) {
        throw std::invalid_argument{"division by zero"}; // if std::invalid_argument has constexpr constructor
    }
    return n / d;
}

constexpr auto b = divide(5, 0); // UNCHANGED: still a compilation failure

constexpr std::optional<unsigned> checked_divide(unsigned n, unsigned d) {
    try {
        return divide(n, d);
    } catch (const std::invalid_argument &) {
        return std::nullopt;
    }
}

constexpr auto a = checked_divide(5, 0); // BEFORE: compilation failure
                                         // AFTER: std::nullopt value
```

Constant evaluation violation behavior won't be changed

```
constexpr int throw_if_odd(const unsigned* p) {
    if (*p % 2 == 1) {
        throw 0;
    } else {
        return 1;
    }
}

constexpr int g() {
    try {
        return throw_if_odd(nullptr);
    } catch (...) {
        return 2;
    }
}

static_assert(g() == 2); // UNCHANGED: still an error, not magically okay because dereferencing an int throws
```

Lifetime of exception object must stay inside constant evaluation

```
constexpr auto fail() -> std::exception_ptr {  
    try {  
        throw failure{};  
    } catch (...) {  
        return std::current_exception()  
    }  
}
```

```
constexpr auto stored_exception = fail(); // NOT ALLOWED: no exception object must leak from constant evaluation
```

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